

# Inverse Problems in Geophysics

## Exercise 1: Simple matrix problems

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# Simple matrix problem

Data are subsumed in the *data vector*

$$\mathbf{d} = [d_1, d_2, \dots, d_N]^T$$

consist of true *model response*  $\mathbf{f}(\mathbf{m})$  plus *noise*  $\mathbf{n}$ :  $\mathbf{d} = \mathbf{f}(\mathbf{m}) + \mathbf{n}$

subsurface described by *model vector*  $\mathbf{m} = [m_1, m_2, \dots, m_M]^T$

# Linear problems

$\mathbf{f}(\mathbf{m})$  is linear with respect to  $\mathbf{m} \Rightarrow$  write as matrix-vector product

$$\mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n}$$

## Examples

Gravimetry, Magnetism, Magnetic Resonance, VSP, straight-ray tomography

 So why not  $\mathbf{m} = \mathbf{G}^{-1}\mathbf{d}$  ?

No, because  $\mathbf{G}$  is usually not invertible, not even quadratic.

# Types of inverse problems (naive)

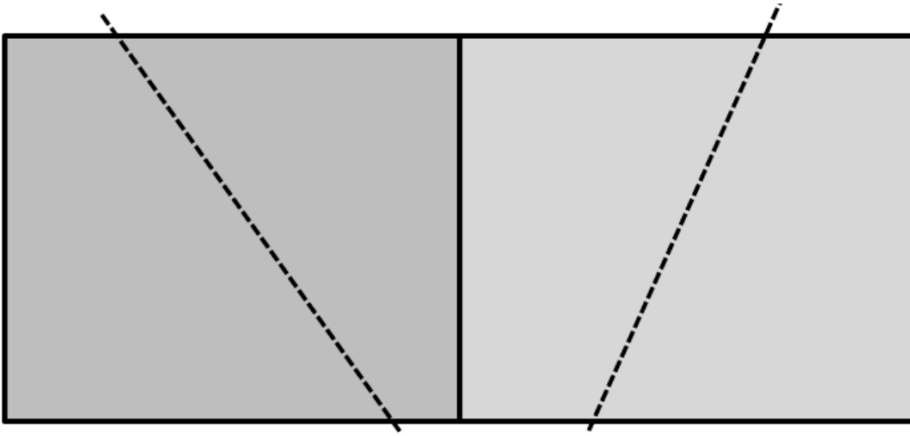
- every row stands for a measurement (data point)
- every column represents a model parameter

$$\mathbf{d} = \mathbf{G}\mathbf{m} + \mathbf{n} \Rightarrow \mathbf{G} \in \mathfrak{R}^{\mathbf{N} \times \mathbf{M}}$$

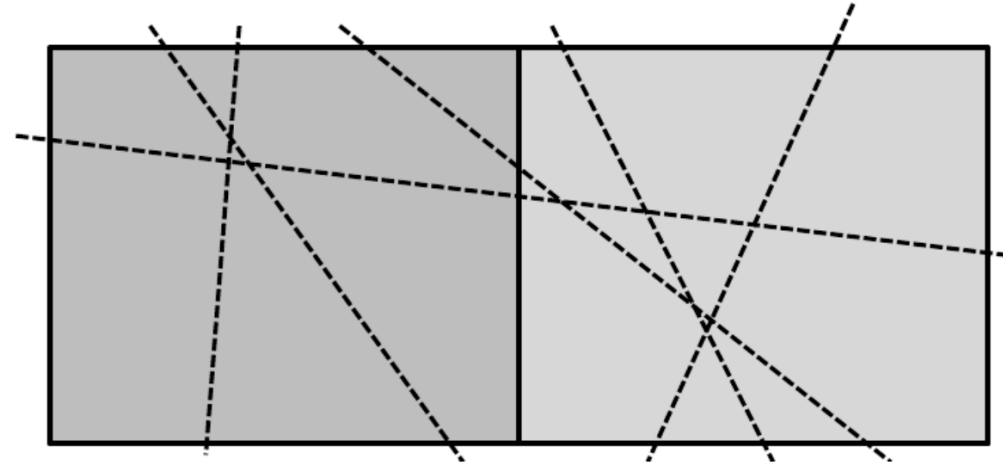
- $N = M$ : even-determined
- $N > M$ : over-determined problem (no existing solution)
- $N < M$ : under-determined problem (no unique solution)
- mostly: (both over- and) under-determined model parts

# Types of inverse problems (Menke, 2012)

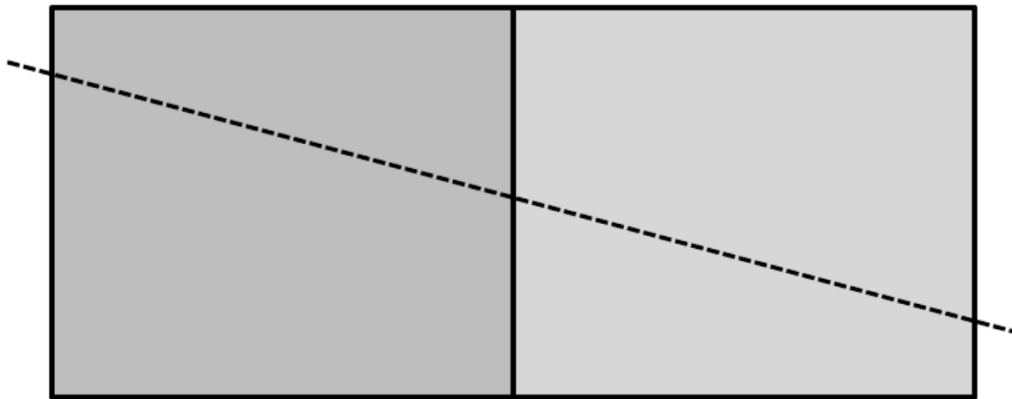
Even-determined



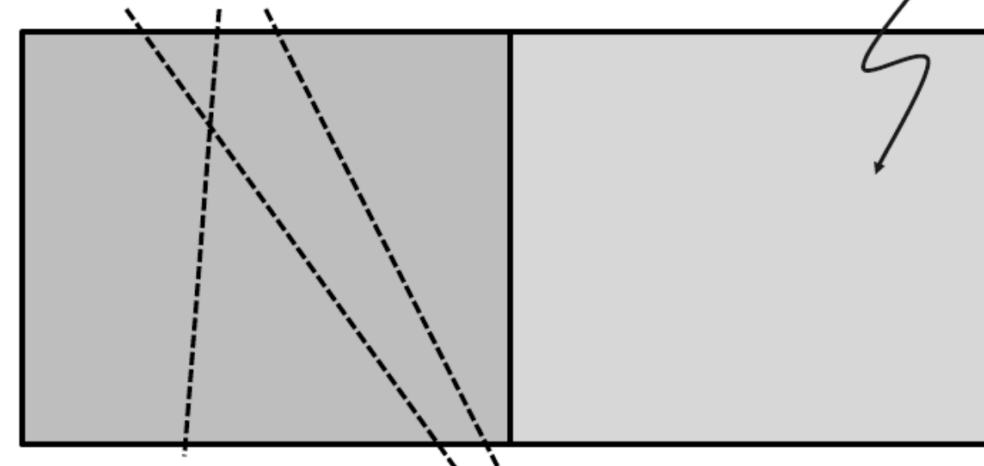
Over-determined



Under-determined



Mixed-determined



# A simple matrix problem

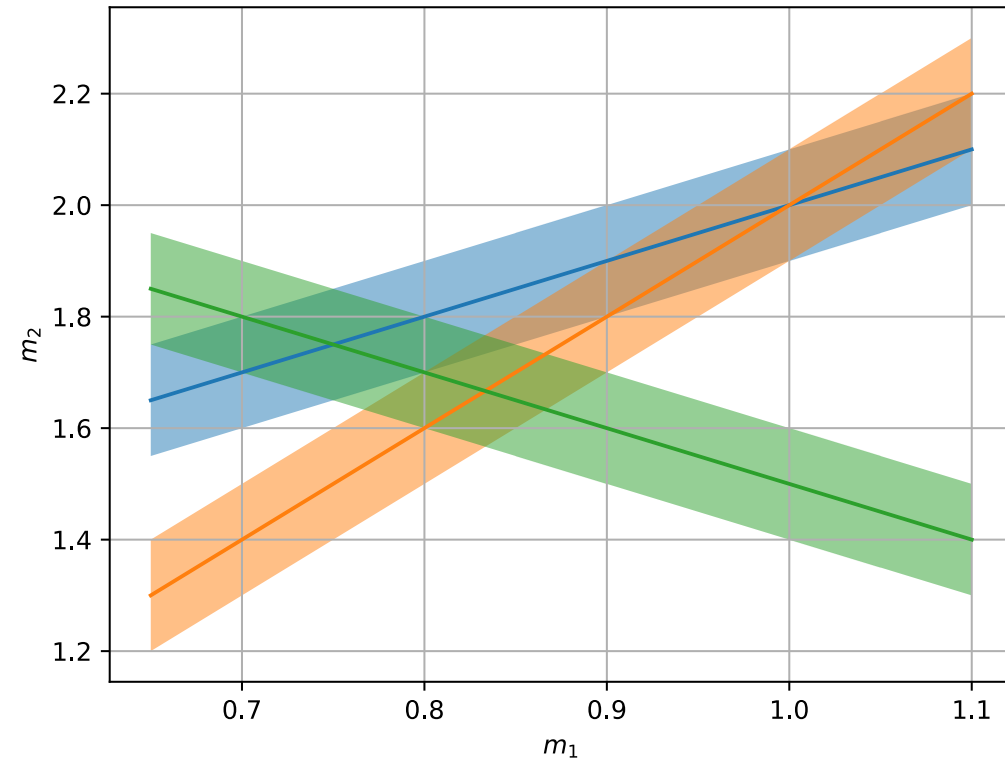
$$m_1 - m_2 = -1$$

$$2m_1 - m_2 = 0$$

$$m_1 + m_2 = 2.5$$

$$\mathbf{G} = \begin{pmatrix} 1 & -1 \\ 2 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{d} = \begin{pmatrix} 1 \\ 0 \\ 2.5 \end{pmatrix}$$

$$m_2 = \frac{d_i - G_{i1} \cdot m_1}{G_{i2}}$$



# Create your Python environment

Install editor of your choice (recommendation: VSCode)

1. The conda way: use simple installer like [MiniForge](#)

```
conda create -n pg -c gimli -c conda-forge pygimli
```

2. The venv/pip way: use environment manager like [UV](#)

```
create virtual enviroment and pip install pygimli
```

# Task

$$m_1 - m_2 = -1$$

$$2m_1 - m_2 = 0$$

$$m_1 + m_2 = 2.5$$

1. Set up the matrix  $\mathbf{G}$  and the data vector  $\mathbf{d}$
2. Reformulate equations into  $m_2(m_1)$
3. Draw the three equations for a range
4. Guess the “best tradeoff”
5. Add error  $(d + \delta d)$  and draw shaded region
6. Increase the error until the regions meet
7. Use a range of values for  $m_1$  &  $m_2$  and perform grid search using objective function



# The objective function

We minimize the L2-norm (summed squared distances) of the residual between data and model response:

$$\Phi = \|\mathbf{d} - \mathbf{G}\mathbf{m}\|_2^2 = \sum_1^N (\mathbf{d}_i - \mathbf{g}_i(\mathbf{m}))^2 \rightarrow \min$$

Let's compute the objective function for a range of values for  $m_1$  and  $m_2$  (grid search).

# Problem types

**Example linear problem: Traveltime tomography**

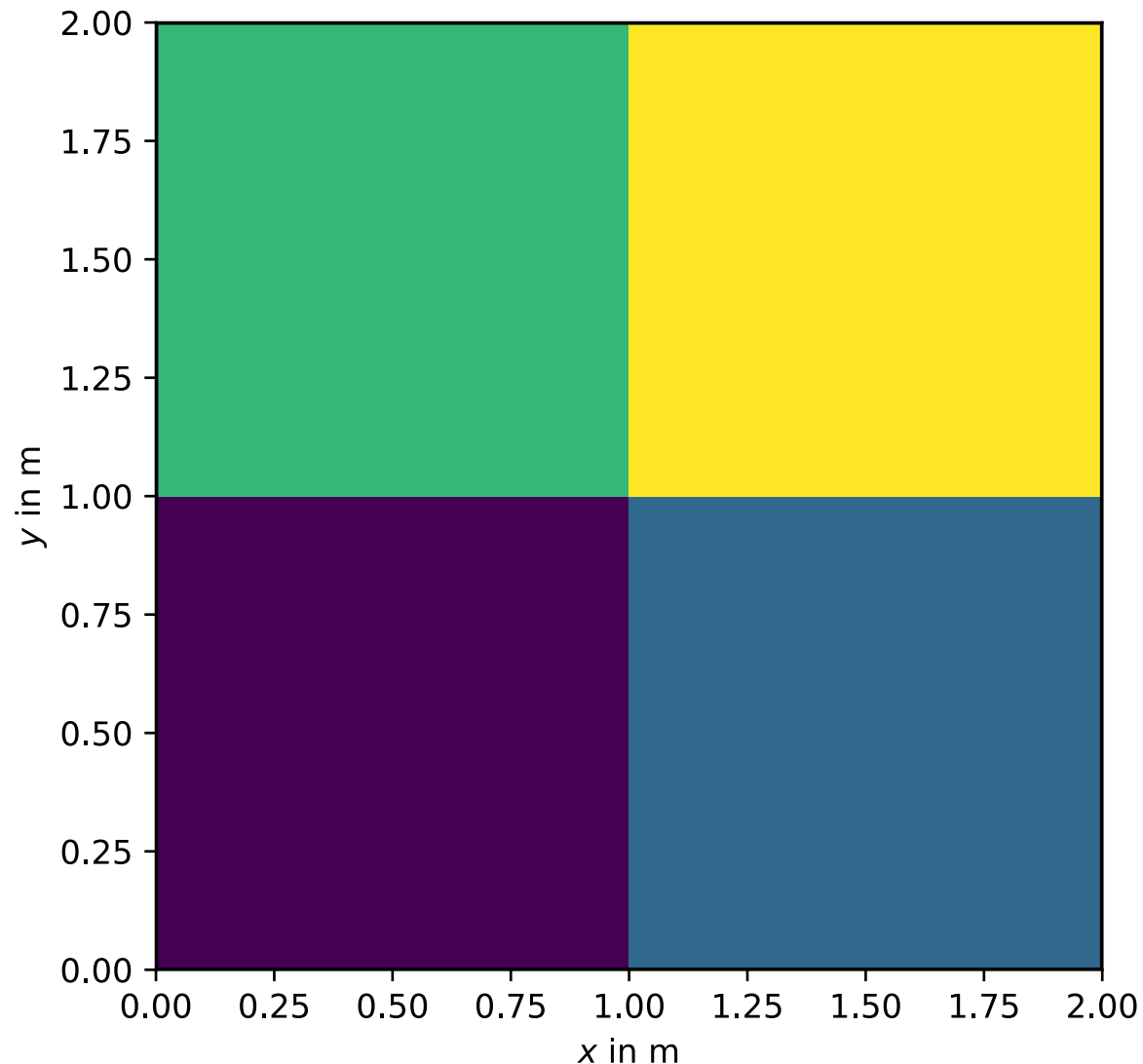
$$t = \int_L \frac{1}{v} dl = \int_L s dl$$

piece-wise constant  $s$ :  $t = \sum_j s_j \Delta l_j \Rightarrow t_i = \sum W_{ij} l_j$

$$\mathbf{d} = \mathbf{W}\mathbf{m}$$

with  $\mathbf{d} = [t_1, \dots, t_N]$ ,  $\mathbf{m} = [s_1, \dots, s_M]$

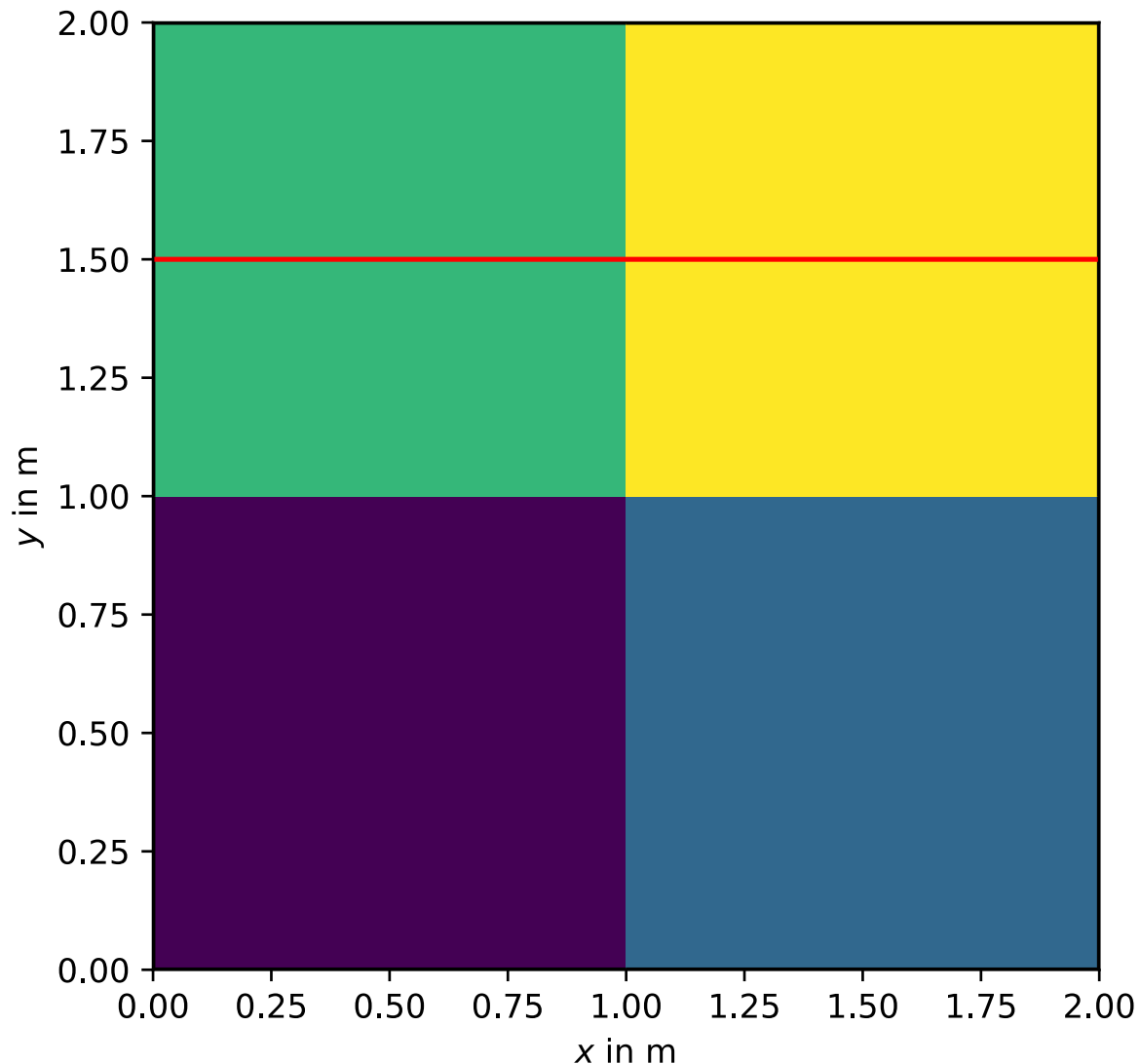
# Problem types - model



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table



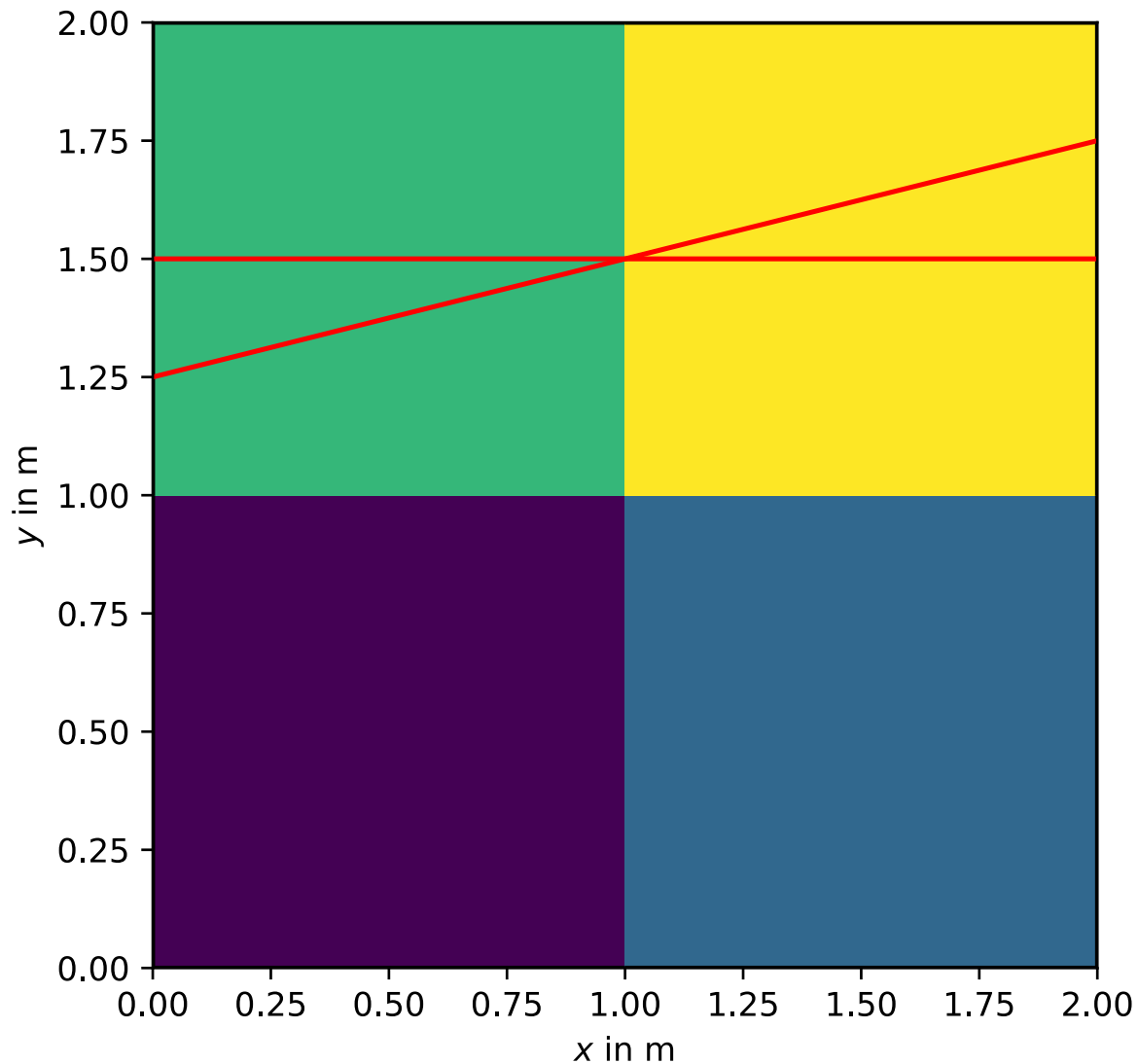
# Problem types - 1



- guess whether it is a) over-, b) under-, or c) mixed-determined
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# Problem types - 2

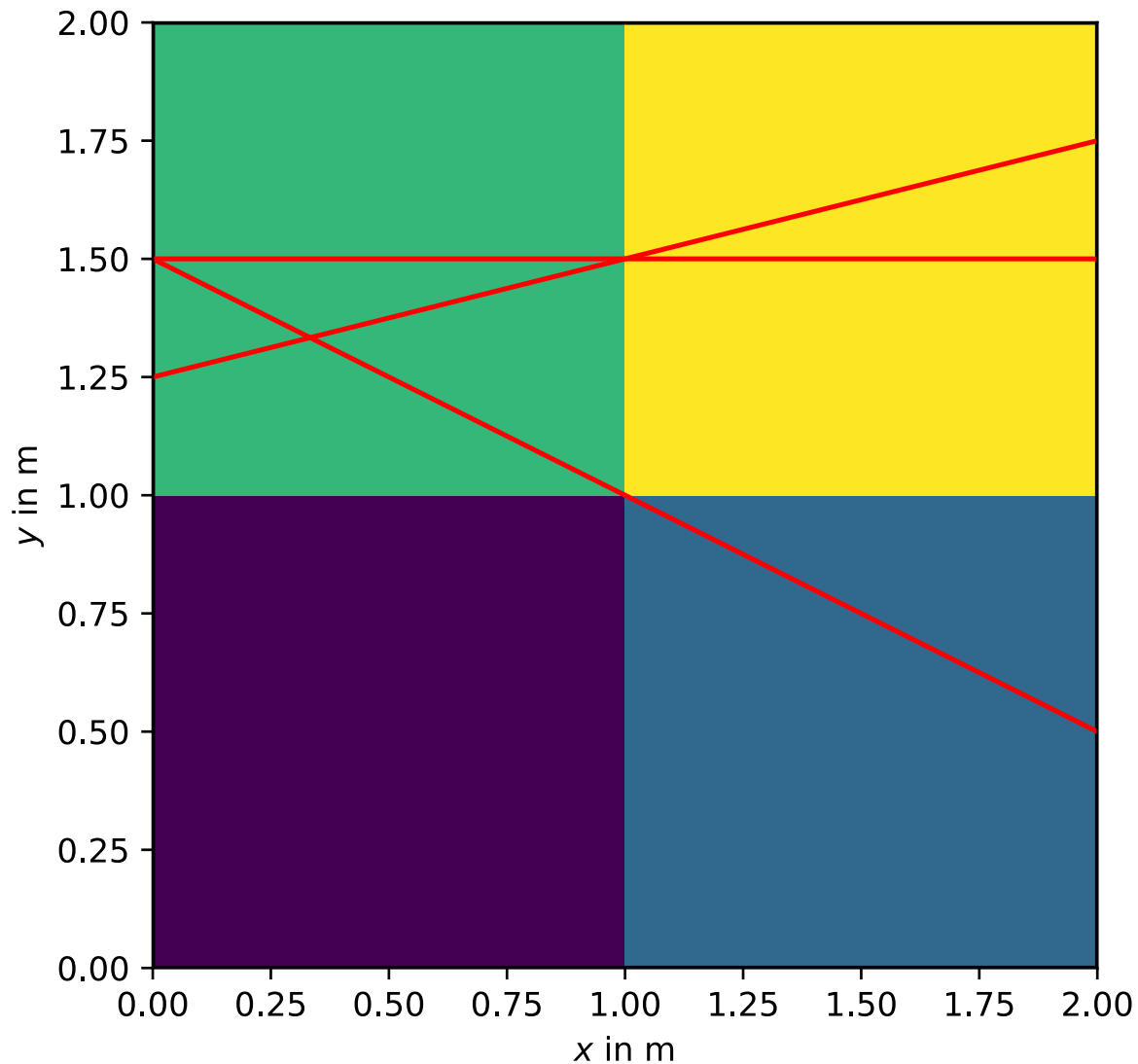


- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
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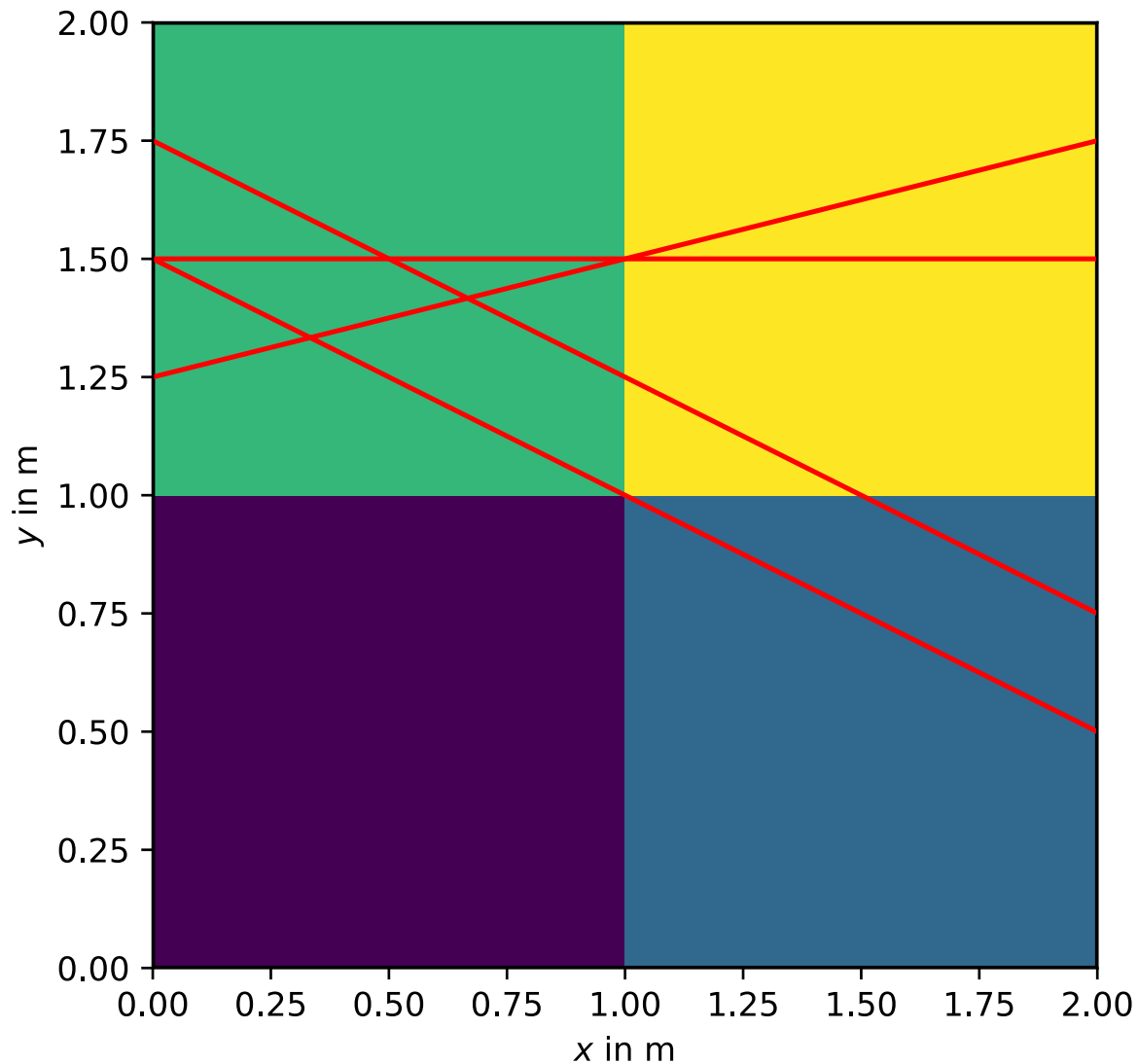
# Problem types - 3



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table



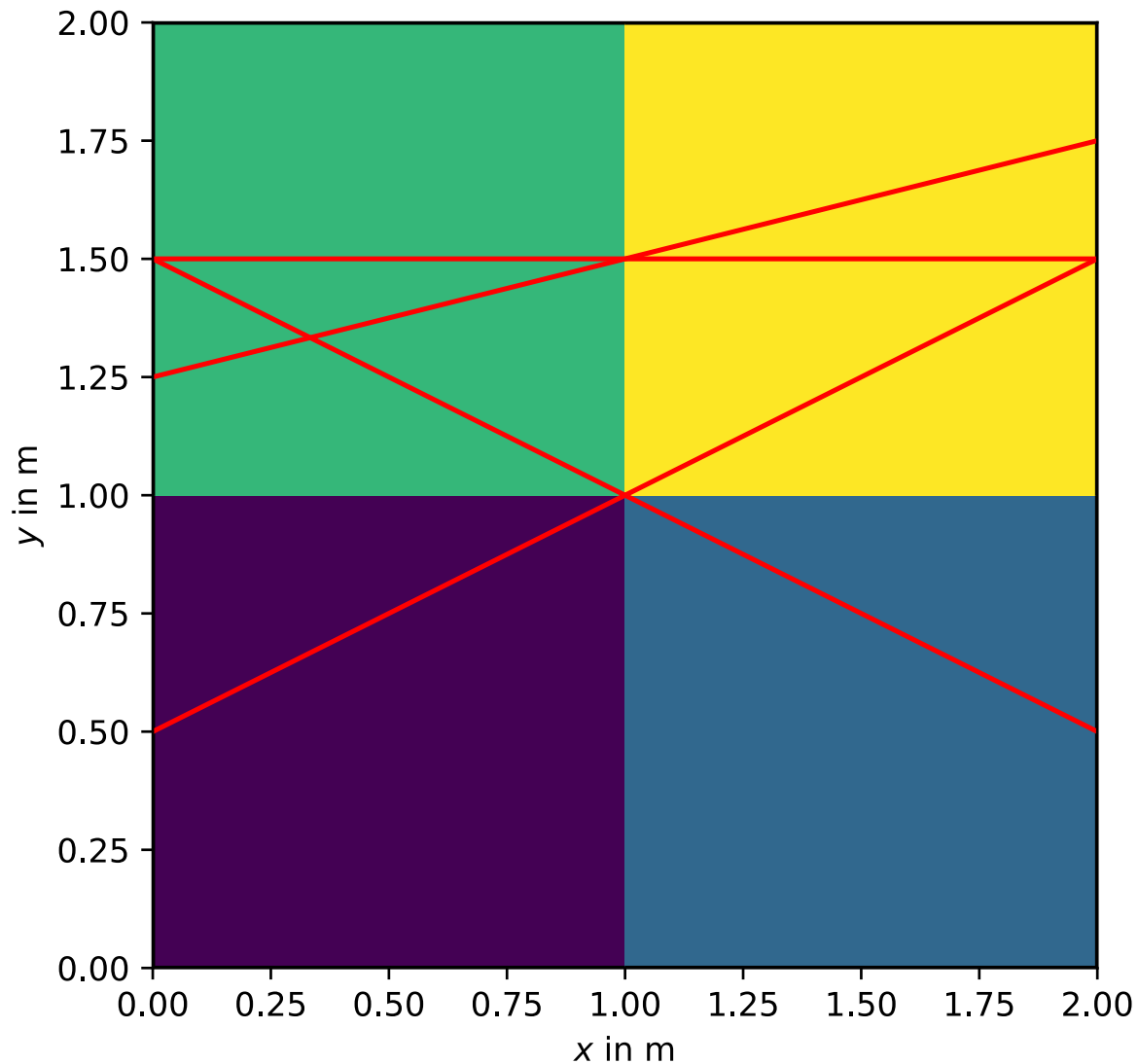
# Problem types - 4



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table



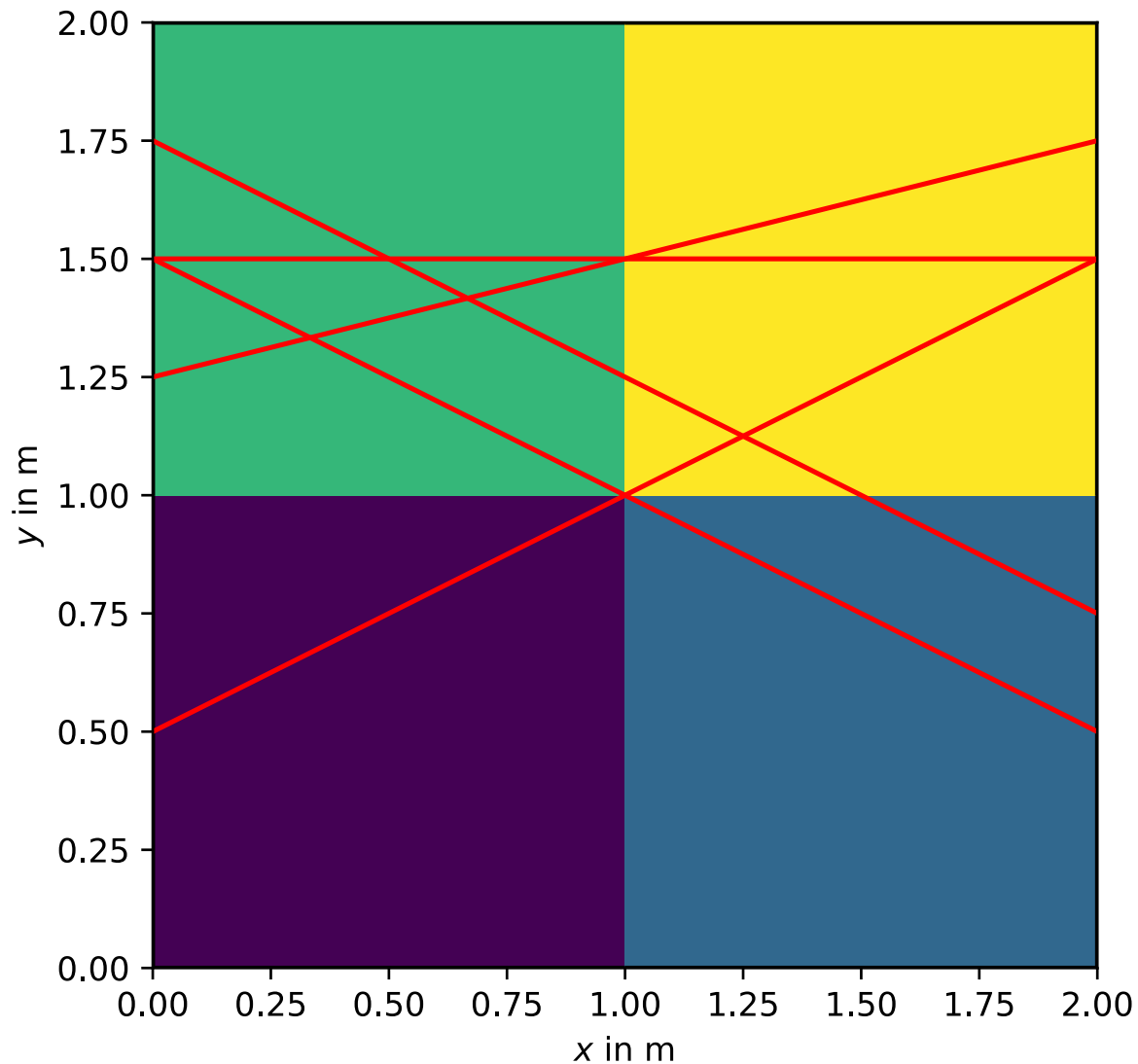
# Problem types - 5



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table



# Problem types - 6

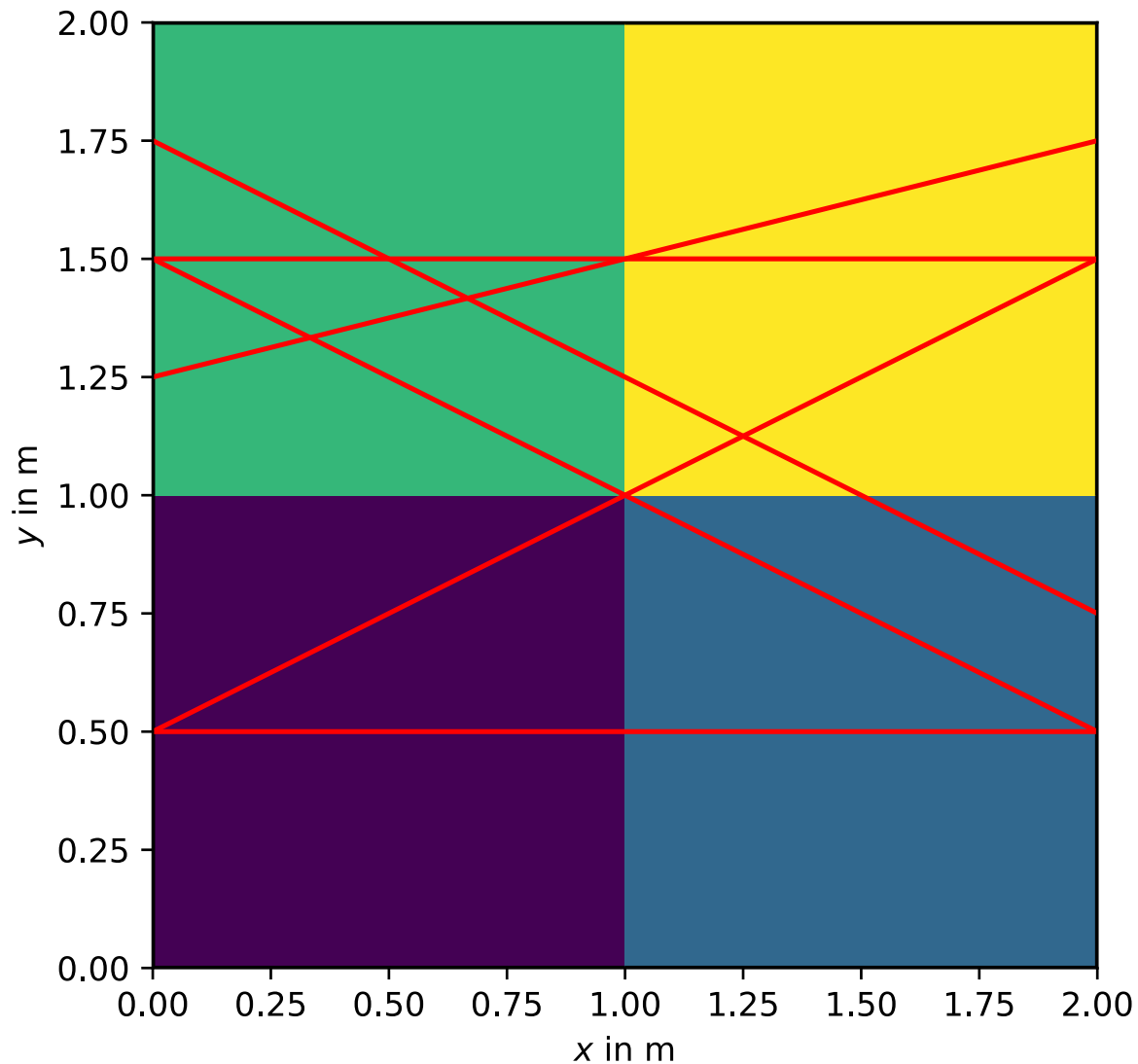


- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table





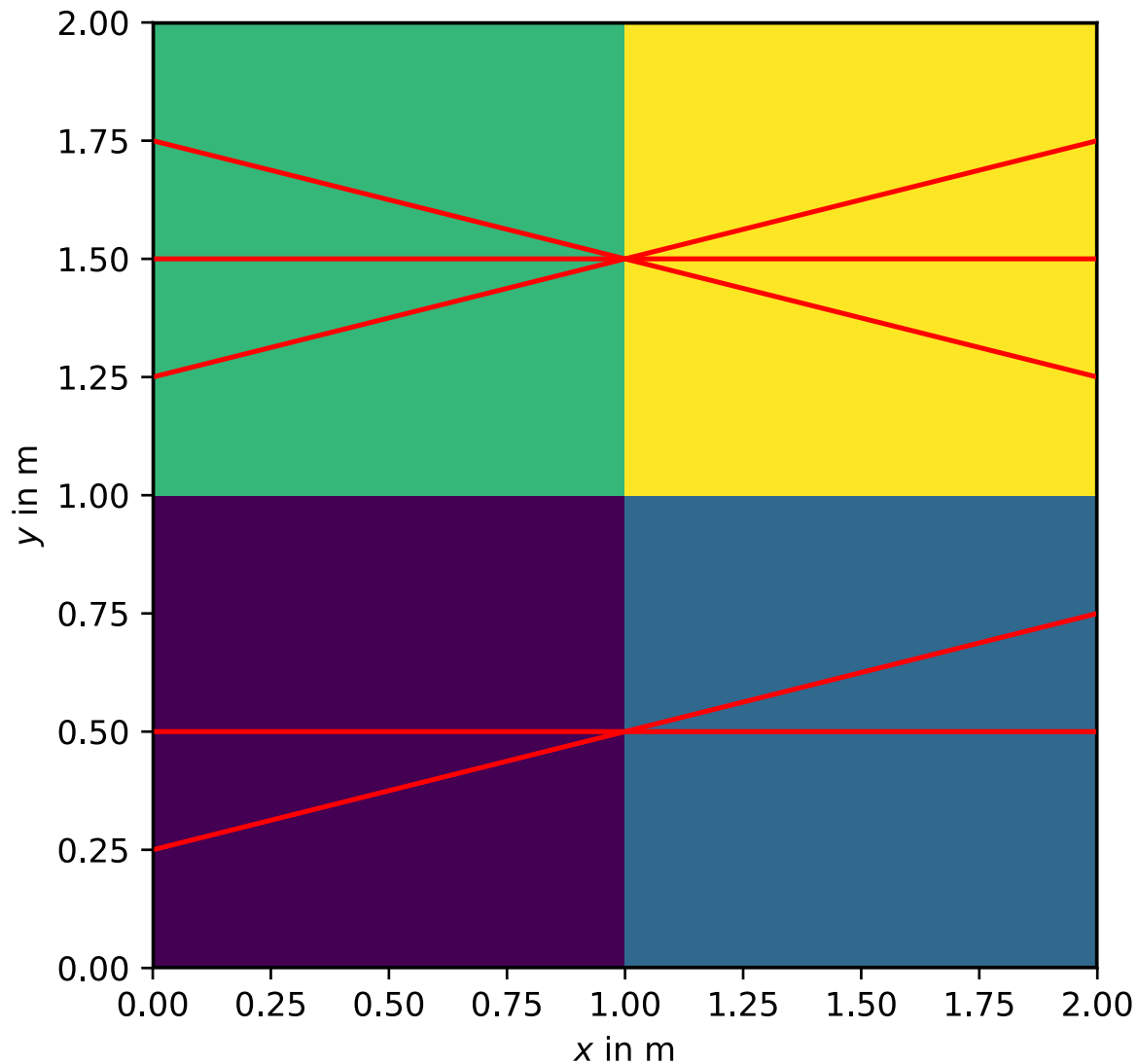
# Problem types - 7



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table



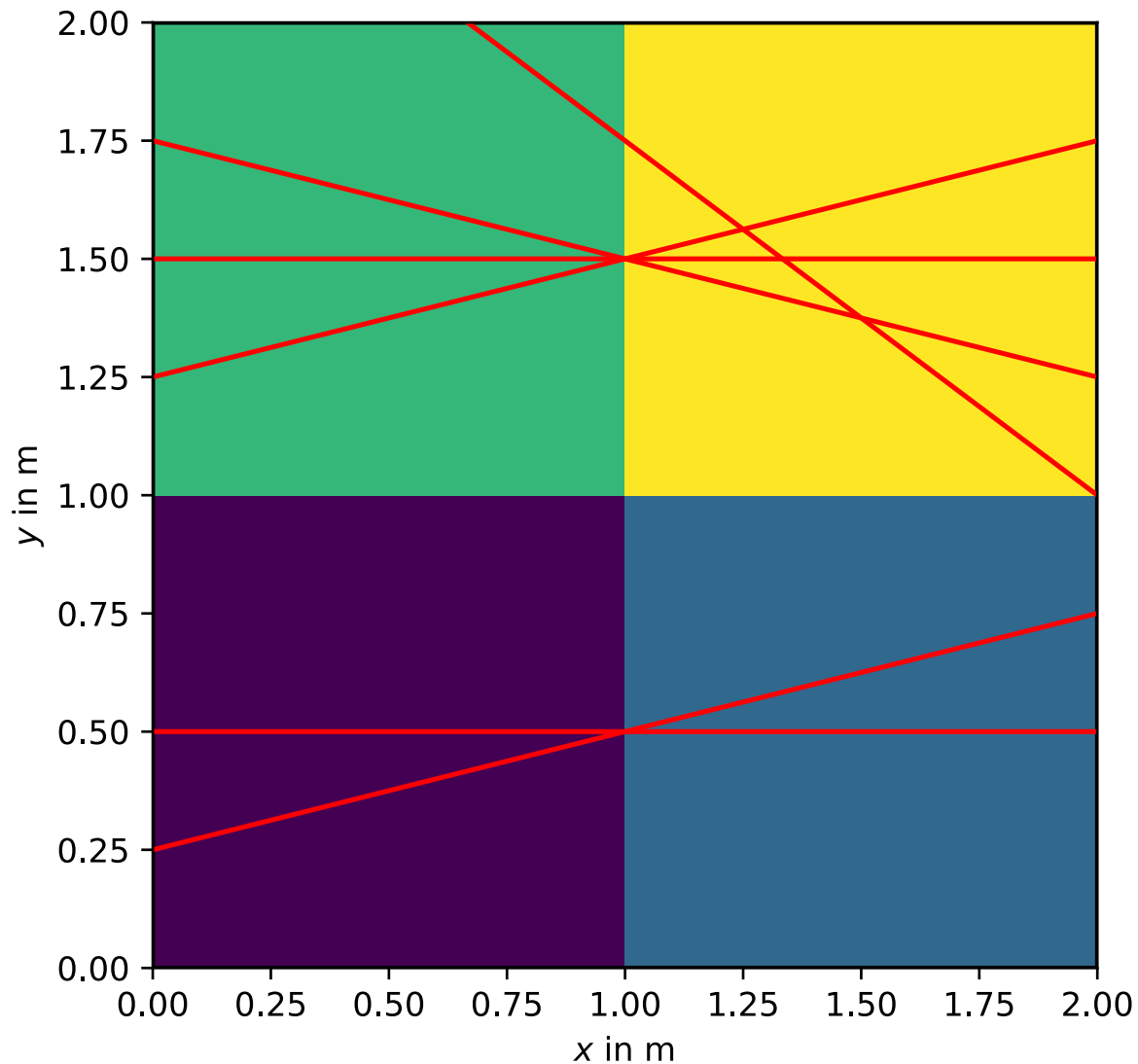
# Problem types - 8



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table



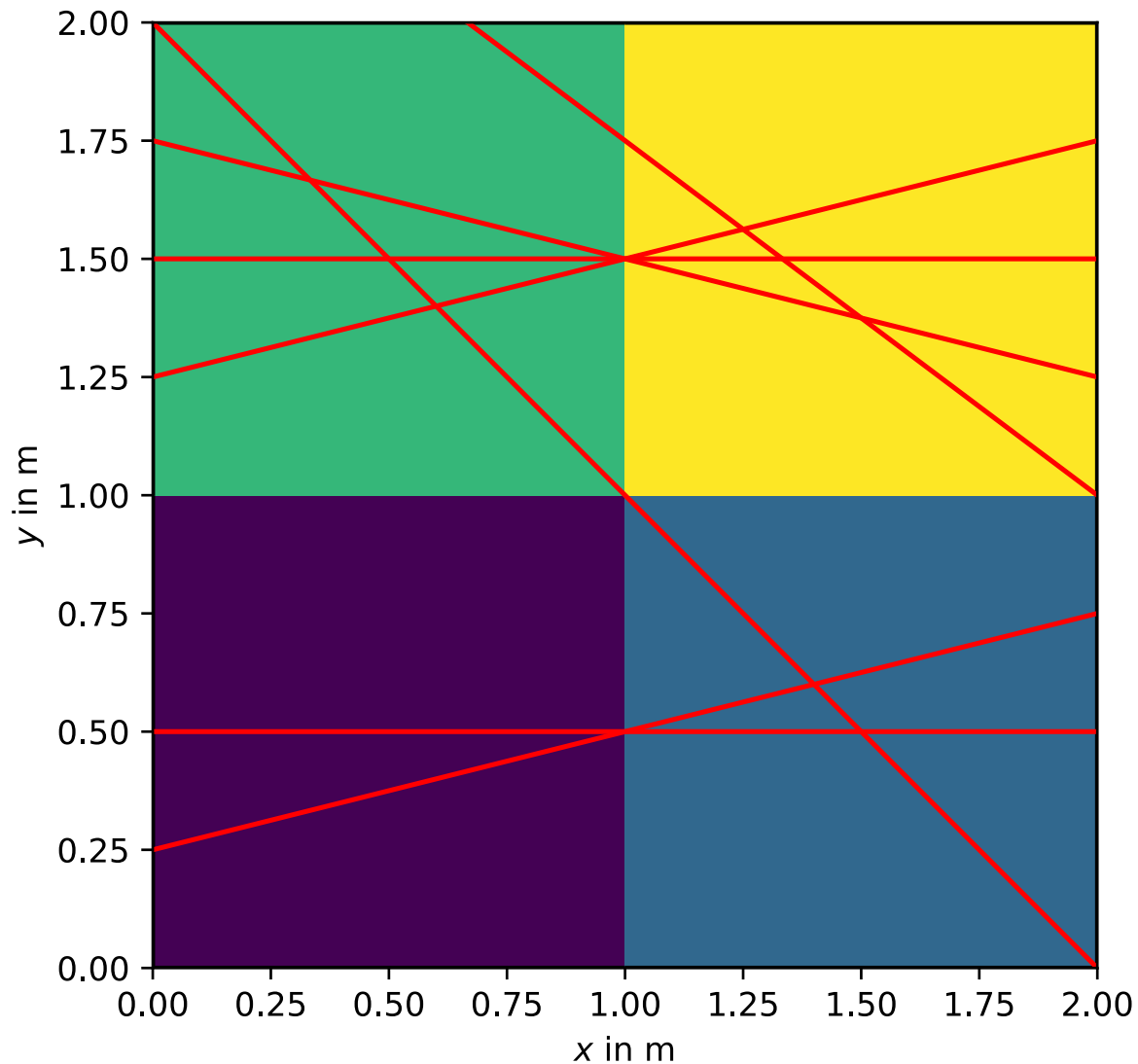
# Problem types - 9



- guess whether it is a) over-, b) under-, or c) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table



# Problem types - 10



- guess a) even, b) over-, c) under-, or d) mixed-determined
- generate the ray path matrix
- determine the rank by using SVD
- generate data from an inhomogeneous model vector
- choose method & compute model
- compare with the backslash
- compile the numbers in a table

